

Diploma in Data
Analysis

Confidence in your Sample

Lesson 5: Summary Notes



Contents

- 3 Lesson 5 objectives
- 3 Introduction
- 4 Population and sample
- 6 Confidence interval and limits
- 8 Data management
- 10 References





Lesson Objectives

- Sample and population
- Confidence interval
- Data management

Lesson Introduction

By the end of this lesson, you will know what the differences between a sample and population and why samples are so important to us. Along with sampling, you will be introduced to confidence intervals. And then we will end the lesson with an interactive data management exercise on our titanic dataset where I will show you a few more tips and tricks to look out for.

Did you know?

There has been an ongoing boom of career opportunities in the last few years. A while ago, it was estimated by Mckinsey that the United States alone faces a shortage of 140 000 - 190 000 employees with analytics expertise. Almost every industry you can think of that uses data, will require data analysts to analyse it.

Population and sample

Imagine we want to gather information about people living in a country. Every 10 years governments must conduct a census which collects information from every person living in that country. As you can imagine this can be very time and resource consuming.

What happens when we want to find information about people in the country quickly? This task would be difficult with an extremely large group. We could conduct a smaller study based on a small proportion of the population. The proportion of the population is representative of the larger group and is known as a sample. We can use this sample to make inferences on the general population of the country.

Population defined

A population can be defined as a large set or collection or group of items. All these items have something in common. An example of a population could be all the students enrolled at Shaw Academy. This would be a group of all students and the students will all have been enrolled at Shaw Academy in common.

Before we begin a study, it is important to know what our population is. We need to define the population to meet the needs of our study. For example, if we want to conduct a study on birth weights between 2 years, we need to define the population as the birth weight of live-born infants in the united states between years x and y. But once again, this task of calculating the mean and variance for this population would be difficult for such a large group.

Sample defined

- A sample can be defined as a subset of the population that is selected to be representative of the larger population. It is a set of data that is collected or selected from a population or larger group through a specific procedure.
- Using the Shaw Academy example, a sample could be all students under the age of 25.

It isn't practical to sample the entire population, therefore we take a sample from that population to deduce information about the larger population. Collecting a smaller group of data is less time consuming and much cheaper.

Sample size

It is important to consider the size of your sample given your population. For example, it's not sensible sampling 3 people when considering the entire population of a country.

If you choose a sample that is too small a representation of your population, the confidence intervals might be very wide, and we increase our risk of obtaining errors in the statistical hypothesis tests.

It is not always possible to sample a bigger sample due to resource restraints.

It is important to remember that larger sample sizes will generally be more accurate in estimating unknown parameters about the population and depending on the precision we want; we select a sample size.

Methods of sampling

The process of collecting information from a sample is called sampling. We use sampling techniques because we want to choose a sample that is a good representation of the population.

A few sampling techniques include:

Random sampling, systematic sampling, stratified sampling, and cluster sampling.

Random sampling

- Random sampling, also known as a probability sample, is when a simple sample size of n , is chosen in a way that every item in the population has an equal chance of being selected to be in the sample.
- A good example would be drawing names randomly from a hat.
- This is the most common way of sampling.
- With simple random sampling, we can use statistical methods to analyse the sample results.

Systematic sampling

- Systematic sampling is when the population is ordered, and items are chosen at regular intervals such as every 5th item in the population.
- The starting point, however, is randomly generated.
- An example of this could be choosing every 10th name in the class starting from the 23rd name in the list.

Stratified sampling

- Stratified sampling is when the population is divided into subgroups, also referred to as strata, based on certain common characteristics, like gender, age, or income bracket.
- For example, if the class has 1500 female students attending but only 300 male students and we want to ensure the sample is an accurate reflection of the population, we could choose a random sample of each group by selecting 150 female students and 30 male students to be a part of our sample.

Cluster sampling

- Cluster sample is when the population is divided into clusters or subgroups.
- The groups all have similar characteristics.
- The subgroups are then randomly selected and all the items in that chosen cluster are selected for the sample.
- This is a good technique for us for large populations that are extremely diverse.

Confidence intervals and limits

Suppose you wanted to know the average number of chips when you order medium fries at McDonald's. If you ordered a few, tallied the total number of chips and divided them by the number of fries ordered you would have obtained what is called a point estimate of the true mean. We can use sample data to make observations about an unknown population. This is called inferential statistics where we use sample data to make an estimate of a population parameter. We know it isn't going to be the exact value, but we can get close to it. We can calculate point estimates from which we can construct interval estimates known as confidence intervals. The confidence interval provides us with a range of values which are likely to contain the population parameter that we are interested in.

Confidence intervals defined

Confidence intervals refer to the probability that a population parameter (like the mean) will be between a set of values for a certain proportion of times. They are used to measure the uncertainty or certainty of a sample method. The most common confidence intervals are 95% and 99%. For instance, we are 95% certain that the mean will lie between these determined values. If we sample the same population many times and achieve a point estimates on each occasion, the confidence interval would bracket the true population parameter in approximately 95% of the cases.

Confidence interval, therefore, contains a range of values that contain the unknown population parameter. The interval provides us with a certainty that the population parameter falls within specific ranges.

Did you know: In the vernacular

You might have heard the phrase "we are 95% certain that group A falls within these ranges". This statement is just another way of saying that the population parameter of interest is contained within the 95% confidence intervals in the vernacular.

Interpreting confidence intervals

Let's use an example to illustrate how we interpret a 95% confidence interval on a normal distribution.

Suppose we have a flock of female sheep happily grazing in a field in France. We want to know what the mean weight of the sheep are. We could take a random sample of 100 sheep from the population and establish a mean weight of 70kg. The mean of 70kg is a point estimate of the population mean (the entire flock of sheep's mean).

We need to know how far this point estimate of the sample is from the true mean of the population. Therefore, we establish a 95% confidence interval using the sample mean and standard deviation and assuming a normal distribution. We find our interval to be between 60 - 80 kg and will arrive at an upper and lower bound that contains the true mean 95% of the time.

This means that if we take 100 random samples from our population of sheep, the means of these should fall in the interval 95% of the time.

We want to be able to say that we are 99% certain that the sample means will fall within a certain interval, meaning we want greater confidence and expand the interval to 99% confidence.

To do so, our confidence interval range will broaden to include a higher number of samples means. For instance, where our previous confidence intervals were between 60-80kg, now the confidence interval will expand to include 60-90kg (for argument's sake). Now, 99 out of 100 times the point estimate of the samples will fall within the confidence interval.

Misconceptions/misunderstandings about confidence intervals

Confidence intervals are often misinterpreted. A confidence interval is not the percentage of data from a given sample that falls within the interval. We cannot say that when we draw a sample from our flock of sheep at a 95% confidence interval of 60 - 90kg, that 95% of the data from the random sample falls between the upper and lower bounds of the interval. To do so, we would need to know the sample mean and standard deviation and be able to identify the sample distribution as normal.



NOTES

Data management

Did you know?

Did you know that analytics professionals can spend up to 60% of their time on data management? Data cleaning is a foundational step to the data analysis process. The step is important to ensure that the results we generate are accurate. When data is drawn from many different sources, this information often carries mistakes.

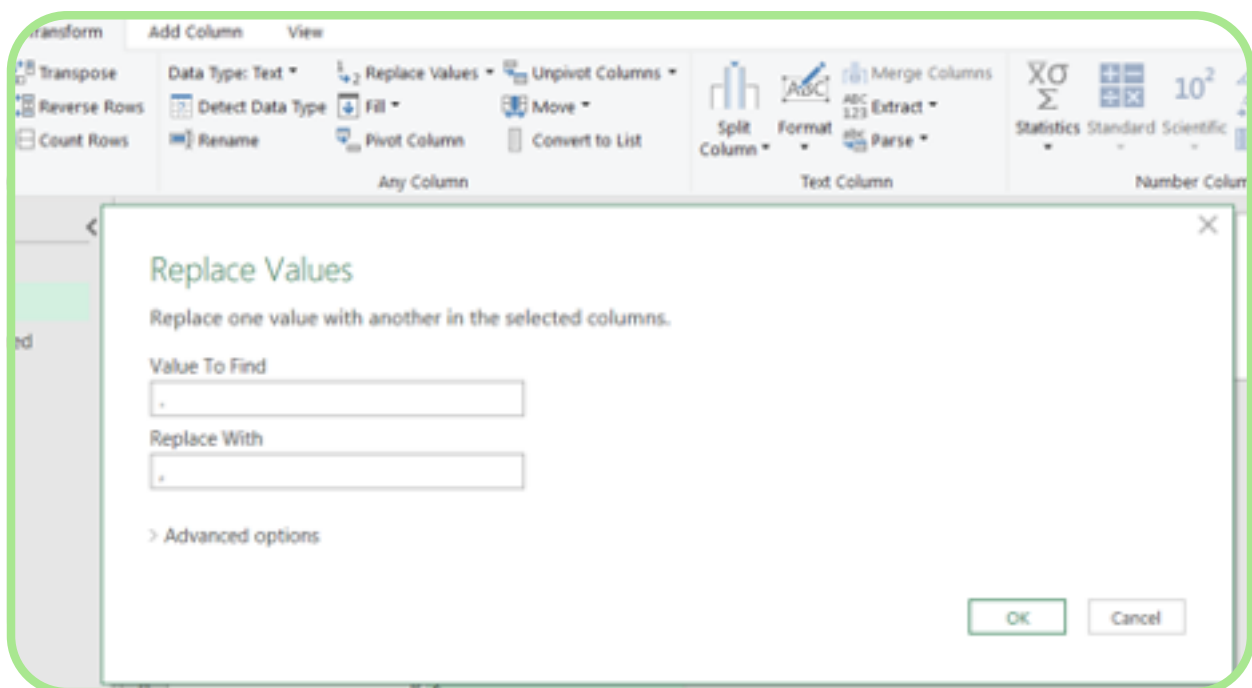
Data management

Data management is important because it optimizes the way that we work, it ensures that the data is accurate and in turn, will increase revenues and efficiency in the working environment.

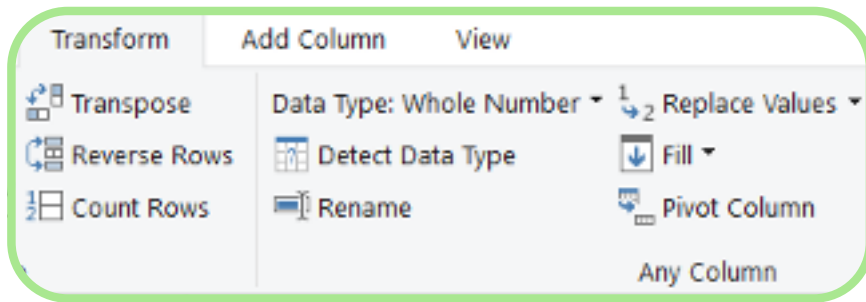
Data cleaning involves removing information not needed, updating information that is incomplete, incorrect or incomplete, checking if there are any duplicate variables, to name but a few of the steps.

Transform a variable

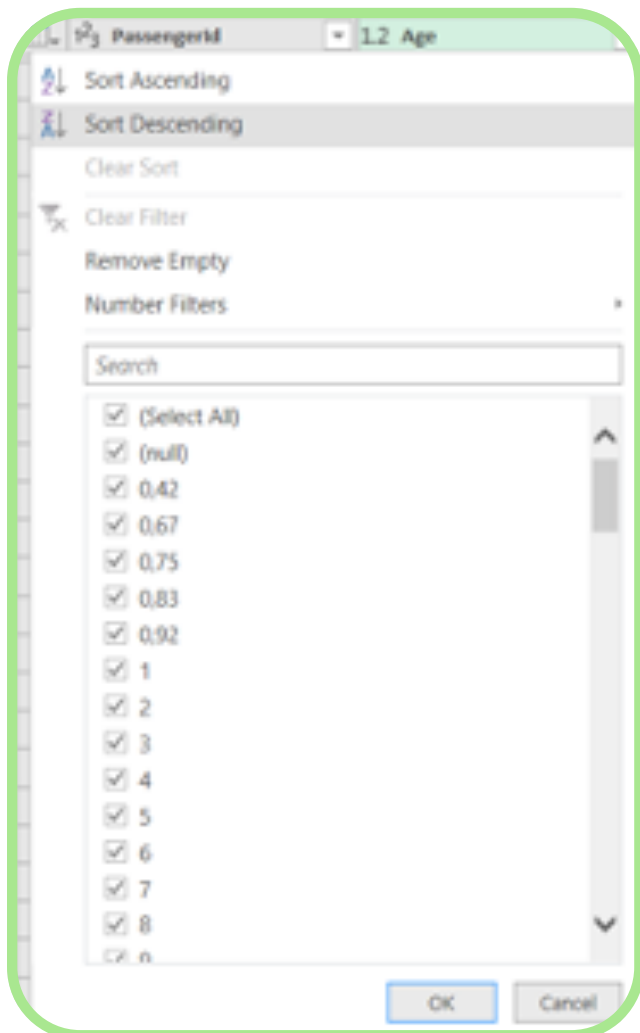
1. Decimal values read in with points can be replaced with a comma.



2. Data type can be changed from, for example, "Whole Number" to "Decimal Number".



3. Null values can be removed from variable by selecting the drop-down and deselecting “(null)”.



Resources

Glen, S., 2013, What is a Population in Statistics?, Statistics How To, <https://www.statisticshowto.com/what-is-a-population/>

365 Data Science, Statistics - Population vs sample, <https://365datascience.com/explainer-video/population-vs-sample/>

Engineering Statistics Handbook, 7.1.4. What are confidence intervals?, 2013, <https://www.itl.nist.gov/div898/handbook/prc/section1/prc14.htm>

Kenton, W., 2020, Confidence Intervals, Investopedia, <https://www.investopedia.com/terms/c/confidenceinterval.asp#:~:text=A%20confidence%20interval%2C%20in%20statistics,certainty%20in%20a%20sampling%20method.>

Yale University: Department of Statistics, 1997, Confidence Intervals, <http://www.stat.yale.edu/Courses/1997-98/101/confint.htm>

Introduction to Confidence Intervals, Lumen course: Introduction to Statistics, <https://courses.lumenlearning.com/introstats1/chapter/introduction-confidence-intervals/>

Langmann, K., How to Calculate a Confidence Interval in Excel, Spreadsheeto, <https://spreadsheeto.com/confidence-interval-excel/>

Description of the CONFIDENCE statistical functions in Excel, Microsoft, <https://support.microsoft.com/en-us/office/description-of-the-confidence-statistical-functions-in-excel-97f5bf0e-5d56-4f8e-8345-2ec1dada8cd5>

Carlberg, C., 2011, Statistical Analysis with Excel 2010: Using Excel with the Normal Distribution, informIT, <https://www.informit.com/articles/article.aspx?p=1717265&seqNum=3>

El-Amir, H., 2019, Titanic (Step 2): Cleaning and Preprocessing, Data is Utopia, <https://dataisutopia.com/blog/preprocessing-titanic-dataset/>

Srivastava, A., 2019, Cleaning the Titanic Dataset [Day 1- #30daysofML], Analytics Vidhya, <https://medium.com/analytics-vidhya/cleaning-the-titanic-dataset-day-1-30daysofml-5cc19294176b>

Sisense, Data Cleaning, <https://www.sisense.com/glossary/data-cleaning/>

Lewandowski, P., 2018, What is data cleaning and why is it important?, sunscrapers, <https://sunscrapers.com/blog/why-is-clean-data-so-important-for-analytics-and-business-intelligence/>

Microsoft, 2020, List Functions, <https://docs.microsoft.com/en-us/powerquery-m/list-functions>

Bassett, E.E., Bremner, J.M., Jolliffe, I.T., et al., Statistics: Problems and Solutions, 2nd edition, London, 1986, pp. 131-134